

Adriel's Portfolio

Introduction

Hello, I'm Adriel —a Software Engineer with three years of well-rounded full-time software-development experience. I excel at designing and implementing end-to-end solutions that leverage cutting-edge technologies to solve real-world problems and deliver tangible value to users.

I hold a bachelor's degree in Bioengineering from Yonsei University and ~2 years of internship experience as a Software Engineer, prior to my full-time employment.

Below are some of the projects I worked on that highlight my skills:

1. Real-Time and Near real-time Transcription (B2C, Chrome Extension and Desktop App, Slid)

Real time transcription of Youtube videos through a chrome extension, and through a desktop application (windows). The resulting transcript is passed through an LLM to generate smart notes as the user watches a video. My contribution was implementing real time transcription on a windows desktop application and maintaining the overall backend system.

Tech Stack:

- Javascript, Electron, WebSockets, Cloud Speech API, Python, Django, Claude Haiku

2. Video to Article Pipeline(3PRO)

Developed an automated AI pipeline that detects a new video on a customer's Youtube channel, and uses AI to generate a natural blog article that covers the

content of the video. The html article is then sent to the customer's server via a webhook for publishing to a premium website.

Tech Stack:

- AWS Eventbridge, Daglo, o1, Python, FastAPI, PostgreSQL

3. RAG-based Chatbot (Investor Advisor Chatbot, 3PRO)

3PRO is a Korean media company specializing in finance. For this B2B project, I developed an AI chatbot that continuously monitors the 3PRO YouTube channels for new videos, automatically transcribes them, and stores the transcripts in a vector database. Using Retrieval-Augmented Generation (RAG), the chatbot can answer user questions by retrieving relevant recent information from the accumulated transcripts and referencing the original video content.

Tech Stack:

- Typescript, NextJs, Python, FastAPI, Pinecone, OpenAI Embeddings, Whisper, Claude Sonnet, AWS Eventbridge, PostgreSQL

4. Course Manuscript + PPT to article (HUNET)

Hunet e-Learning platform has a large number of video courses. In order to boost SEO the client wanted to make high-quality blog articles from the manuscript and ppt used to prepare the video courses. I developed an Pipeline that receives the manuscript and the ppt (storyboard) and use AI to semantically merge the information and generate a high-quality blog article.

Tech Stack:

- Gemini Flash Vision, GPT4o, Python, DALL-E

5. Summary (B2C Mobile App)

A mobile app, where users subscribes to Youtube channels, and when new videos are uploaded, the app automatically generates a brief TEXT summary of the video and notifies the user. I contributed the system design and backend development.

Tech Stack:

- Typescript, React Native, NestJS, MongoDB, Whisper, GPT4o, AWS Lambda

6. DipClip (B2C Mobile App)

A mobile app, where users subscribes to Youtube channels, and when new videos are uploaded, the app automatically generates bit sized VIDEO summaries and notifies the user. I contributed the system design and backend development.

Tech Stack:

- Typescript, React Native, Python, FastAPI, PostgreSQL, Whisper, GPT4o, AWS Lambda

7. KOHI Content Mind Map (한국보건복지인재원 콘텐츠 마인드맵)

<https://kohi.bebridge.ai/>

Developed KOHI's Content Mind Map—a web application designed to help users discover relevant video courses from a library of 100+ offerings. I used AI to auto-transcribe each course, generate summaries, and extract hierarchical topic metadata. The application's interface then visualizes these topics, enabling seamless navigation and course discovery.

Tech Stack:

- Typescript, NextJs, Python, FastAPI, Pinecone, Daglo, O1

8. Multi-Language Video Dubbing Pipeline (Bebridge AI)

At Bebridge, we built an AI-powered video dubbing studio to automate multilingual localization while preserving the original speaker's voice. I led the design and development of **Part 1** of the pipeline, which includes:

- Automatic transcription of the original video
- AI Agent translation adapted to match the timing and rhythm of the source audio
- Voice cloning for speaker consistency across languages
- AI-generated speech synthesis aligned with the original audio duration and tone

This system enables fast, high-quality, voice-preserving dubbing.

Tech Stack:

- Cloud Run, Python, FastAPI, PostgreSQL, Whisper, Daglo, Speechmatics, AI Agents